
Bridging the Substrate Gap: Architectural Patterns for Unified Audio-Text Multimodal Intelligence

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Abstract

The *substrate gap* between text tokens and audio tokens constitutes the core engineering challenge for native real-time multimodal artificial intelligence. Text tokens, produced by byte-pair encoding at 3–5 tokens per second of speech, are purely semantic, discrete, and low-temporal-resolution. Audio tokens, produced by neural codecs at 50–400+ frames per second, entangle semantics, emotion, prosody, and timbre within dense, high-temporal-resolution representations. Bridging these two substrates—so that a single model can listen, reason, and speak with sub-200 ms latency and zero context loss—requires dissolving fundamental mismatches in vocabulary, temporal resolution, routing, and paralinguistic preservation. We survey the four principal architectural patterns that address these mismatches: (1) *discrete neural codecs* for vocabulary unification, (2) *semantic-acoustic factorization* for temporal alignment, (3) *shared KV-cache routing* for zero-latency modality handoff, and (4) *interleaved style tokens* for paralinguistic preservation. We provide a unified taxonomy grounded in analysis of five deployed or near-deployed systems—GPT-4o, Moshi, Spirit LM, GLM-4-Voice, and SpeechGPT—and present a case study of a multi-agent sidecar architecture that bridges the gap across system boundaries using NVIDIA PersonaPlex, a frontier reasoning LLM, and persistent memory. We identify open research directions including multilingual codec transfer, real-time emotion adaptation, and the ethical implications of voice cloning at scale.

1 Introduction

The dominant paradigm for voice-enabled large language models (LLMs) is the *cascaded pipeline*: an automatic speech recognition (ASR) module transcribes user audio into text, a language model generates a textual response, and a text-to-speech (TTS) module synthesizes the reply Radford et al. [2023], Shen et al. [2018]. While this architecture leverages mature components, it suffers from three compounding limitations:

1. **Latency accumulation.** Each stage introduces processing delay. End-to-end latency in production cascaded systems typically exceeds 1–3 seconds, far above the ~ 200 ms floor of natural human turn-taking Dong and Xu [2019].
2. **Context loss at module boundaries.** The ASR transcript discards prosodic, emotional, and speaker-identity information. The TTS module must reconstruct these cues from text alone, a fundamentally ill-posed problem Schuller and Batliner [2013].
3. **Paralinguistic destruction.** A whispered, hesitant question and a shouted, confident one produce identical transcripts. Cascaded systems are structurally incapable of preserving—let alone reasoning about—these distinctions.

The alternative is *native multimodal* modeling: a single Transformer backbone that ingests and produces both text and audio tokens, maintaining a unified internal representation across modalities. Recent systems—GPT-4o OpenAI [2024], Moshi Défossez et al. [2025], Spirit LM Nguyen et al. [2024], GLM-4-Voice Zeng et al. [2024]—demonstrate the feasibility of this approach, but they confront a shared obstacle that we term the **substrate gap**.

The substrate gap. Text and audio inhabit fundamentally different representational substrates. Text tokens are produced by byte-pair encoding (BPE) Sennrich et al. [2016]: a vocabulary of $\sim 100\text{K}$ discrete symbols encoding semantic content at roughly 3–5 tokens per second of equivalent speech. Audio tokens emerge from neural codecs Défossez et al. [2023], Zeghidour et al. [2022]: 1,024–4,096 discrete codebook entries per quantization level, operating at 50–400+ frames per second, and *entangling* semantic content with pitch, timbre, speaking rate, emotion, and acoustic environment. The mismatch is not merely quantitative—it is structural. Three seconds of conversational speech compresses to roughly 10 BPE tokens but expands to 150–1,200 audio tokens, depending on codec configuration. A Transformer attending over both modalities must reconcile two streams with temporal resolution ratios of 30:1 to 100:1, fundamentally different information densities, and no natural alignment boundary.

Contributions. This paper provides:

1. A **unified taxonomy** of four architectural bridges—vocabulary unification, temporal alignment, routing, and paralinguistic preservation—that collectively dissolve the substrate gap (§2–§5).
2. A **comparative systems analysis** of five deployed or near-deployed native multimodal speech-language models (§6).
3. A **case study** of a multi-agent sidecar architecture that bridges the gap across system boundaries, including a novel token-injection mechanism for content delivery through a Speech LM (§7).
4. Identification of **open research directions** and ethical considerations (§8).

2 Bridge I: Unifying the Representational Space

The first bridge converts raw audio waveforms into discrete or continuous tokens that can be processed alongside text within a shared vocabulary. We distinguish three strategies, each with distinct tradeoffs in information fidelity, vocabulary overhead, and training complexity.

2.1 Discrete Neural Codecs

Neural audio codecs Défossez et al. [2023], Zeghidour et al. [2022], Défossez et al. [2024] compress raw waveforms into sequences of discrete integer codes via *residual vector quantization* (RVQ). Given an input waveform $\mathbf{x} \in \mathbb{R}^{T \cdot f_s}$ at sample rate f_s , an encoder E produces a latent sequence $\mathbf{z} = E(\mathbf{x}) \in \mathbb{R}^{T' \times d}$, where $T' = T \cdot f_s / r$ for stride r . RVQ then applies Q successive quantization stages:

$$\hat{\mathbf{z}}_q = \sum_{i=1}^Q \text{VQ}_i \left(\mathbf{z} - \sum_{j=1}^{i-1} \hat{\mathbf{z}}_j \right), \quad \text{VQ}_i(\mathbf{r}) = \arg \min_{\mathbf{c} \in \mathcal{C}_i} \|\mathbf{r} - \mathbf{c}\|_2 \quad (1)$$

where $\mathcal{C}_i = \{\mathbf{c}_1^{(i)}, \dots, \mathbf{c}_K^{(i)}\}$ is the codebook for stage i with K entries (typically $K = 1024$). Each quantization stage captures progressively finer residual detail: the first codebook encodes coarse spectral shape and phonetic content; later codebooks add timbre, room acoustics, and micro-prosodic variation.

Key codecs. **EnCodec** Défossez et al. [2023] operates at 24 kHz with configurable bitrates (1.5–24 kbps), producing 75 frames/sec with up to 32 RVQ levels. **SoundStream** Zeghidour et al. [2022] uses a similar architecture but with an additional discriminator loss for perceptual quality. **Mimi** Défossez et al. [2024] introduces *semantic-acoustic disentanglement* within the codec itself, splitting the first RVQ level into a semantic stream (aligned to HuBERT clusters) and subsequent levels into an acoustic stream. This factorization—which we revisit in §3—is architecturally consequential because it provides a natural alignment point between audio and text tokens.

Table 1: Comparison of vocabulary unification strategies for audio-text multimodal models.

Strategy	Representation	Vocab Overhead	Generation	Info. Loss
Joint discrete (RVQ)	Integer codes	$+K \cdot Q$ tokens	Autoregressive	Quantization
Interleaved discrete	Integer + style codes	$+K \cdot Q + S$	Autoregressive	Moderate
Continuous projection	Dense vectors	None (adapter params)	Requires decoder head	Minimal
Hybrid (Mimi-style)	Semantic discrete + acoustic continuous	$+K$ tokens	Mixed	Low

2.2 Joint Vocabulary Expansion

The most direct unification strategy expands the LLM’s vocabulary to include audio codebook tokens. Let $\mathcal{V}_{\text{text}}$ denote the original BPE vocabulary ($|\mathcal{V}_{\text{text}}| \approx 100\text{K}$) and $\mathcal{V}_{\text{audio}} = \{a_{k,q} : k \in [K], q \in [Q]\}$ the audio codebook indices across Q RVQ levels. The joint vocabulary is:

$$\mathcal{V}_{\text{joint}} = \mathcal{V}_{\text{text}} \cup \mathcal{V}_{\text{audio}}, \quad |\mathcal{V}_{\text{joint}}| = |\mathcal{V}_{\text{text}}| + K \cdot Q \quad (2)$$

For $K = 1024$ and $Q = 8$, this adds 8,192 tokens—a modest expansion for modern LLMs. The model is then trained (or fine-tuned) on interleaved text–audio sequences using standard next-token prediction:

$$\mathcal{L}_{\text{NTP}} = - \sum_{t=1}^N \log p_{\theta}(v_t \mid v_1, \dots, v_{t-1}), \quad v_t \in \mathcal{V}_{\text{joint}} \quad (3)$$

Spirit LM Nguyen et al. [2024] and SpeechGPT Zhang et al. [2023] adopt this strategy, prepending special modality tokens ([TEXT], [AUDIO]) to signal transitions. The advantage is architectural simplicity: the Transformer backbone, attention mechanism, and sampling procedure remain unchanged. The disadvantage is that the model must learn cross-modal alignment purely from data co-occurrence, without explicit structural guidance.

2.3 Continuous Latent Projection

An alternative to discrete vocabulary expansion projects continuous audio representations into the LLM’s embedding space via learned adapters. The **Perceiver Resampler** Jaegle et al. [2021] and **Q-Former** Li et al. [2023] are representative:

$$\mathbf{H}_{\text{audio}} = \text{CrossAttn}(\mathbf{Q}_{\text{learnable}}, \mathbf{Z}_{\text{encoder}}; \mathbf{W}_{\text{proj}}) \in \mathbb{R}^{M \times d_{\text{model}}} \quad (4)$$

where $\mathbf{Q}_{\text{learnable}} \in \mathbb{R}^{M \times d_q}$ is a set of M learnable query vectors, $\mathbf{Z}_{\text{encoder}}$ is the output of a frozen audio encoder, and $\mathbf{W}_{\text{proj}}$ maps to the LLM’s hidden dimension d_{model} . This produces M soft audio tokens—typically $M \ll T'$, achieving simultaneous dimensionality reduction and modality alignment.

Tradeoffs. Table 1 summarizes the design space. Discrete approaches preserve the autoregressive generation interface but quantize away fine-grained acoustic detail. Continuous approaches retain more information but complicate generation: the model cannot autoregressively sample continuous vectors without an additional diffusion or flow-matching head.

3 Bridge II: Solving the Temporal Mismatch

Even after vocabulary unification, a fundamental temporal mismatch remains. Text tokens encode semantic content at roughly 3–5 tokens per second of equivalent speech. Audio codec tokens operate at 12.5–75 frames per second per RVQ level. Raw audio features (mel spectrograms, wav2vec2 outputs) produce 50–100 frames per second. A Transformer processing a 30-second conversation segment must attend over ~ 100 text tokens alongside $\sim 2,250$ audio tokens (at 75 Hz)—a 22:1 ratio that wastes context window capacity and dilutes attention.

3.1 Semantic–Acoustic Factorization

The most architecturally principled solution splits audio into two streams with distinct temporal resolutions and information content:

1. **Semantic stream** ($\sim 25\text{--}50$ Hz): Extracted by self-supervised models such as HuBERT Hsu et al. [2021] or w2v-BERT Chung et al. [2021], then discretized via k -means clustering. These tokens capture phonetic and lexical content—they are *approximately alignable* with text tokens because they encode similar information at a higher but compressible temporal resolution.
2. **Acoustic stream** ($\sim 50\text{--}75$ Hz): The residual after semantic extraction, encoding pitch contour (F_0), energy dynamics, spectral tilt, speaker identity, and room characteristics. These tokens are *not* alignable with text and must be processed through a parallel or delayed pathway.

AudioLM Borsos et al. [2023] pioneered this factorization for unconditional audio generation. Moshi Défossez et al. [2025] extends it to dialogue by using the Mimi codec’s built-in semantic–acoustic split: the first RVQ level is distilled to align with HuBERT semantic clusters, while levels 2–8 encode acoustic residuals. This allows the language model to “reason” over the semantic stream at a manageable temporal resolution while a lightweight acoustic decoder reconstructs full-fidelity audio from the combined streams.

3.2 Temporal Downsampling

Where semantic–acoustic factorization provides a conceptually clean separation, temporal downsampling offers a more direct engineering solution: compress n consecutive audio frames into a single token. Common approaches include:

Strided 1D convolutions. A convolutional layer with stride s reduces the temporal resolution by a factor of s . EnCodec uses successive strided convolutions in its encoder ($s \in \{8, 5, 4, 2\}$ for a total downsampling of $320\times$), achieving 75 frames/sec from 24 kHz audio.

Temporal attention pooling. A cross-attention layer with $\lfloor T'/n \rfloor$ learnable query vectors pools every n frames into one:

$$\mathbf{h}_i = \text{Attn}(\mathbf{q}_i, \{\mathbf{z}_{(i-1)n+1}, \dots, \mathbf{z}_{in}\}), \quad i = 1, \dots, \lfloor T'/n \rfloor \quad (5)$$

This is differentiable and content-adaptive: a pool over silence can compress aggressively, while a pool over a phoneme boundary may retain more detail.

3.3 Compression Ratio Analysis

The practical consequence of temporal mismatch is context window consumption. Consider a 30-second utterance at 24 kHz:

- **Text (BPE):** $\sim 100\text{--}150$ tokens (assuming ~ 4 words/sec, ~ 1.2 tokens/word).
- **Semantic tokens** (HuBERT, 25 Hz): 750 tokens.
- **Codec tokens** (EnCodec, 75 Hz, 8 levels): $75 \times 30 \times 8 = 18,000$ tokens.
- **Raw mel frames** (100 Hz): 3,000 frames.

For a Transformer with a 32K-token context window, a single 30-second audio segment at full codec resolution consumes over 56% of the context. This compression–capacity tension motivates the framerate pyramid shown in Figure 1, where each level represents a different temporal resolution and the model selects the appropriate level based on processing requirements.

4 Bridge III: Engineering Seamless Handoff

Given unified tokens and aligned temporal resolutions, the remaining challenge is *routing*: how does a single Transformer switch between processing audio input, performing textual reasoning, and generating audio output without incurring latency at modality boundaries?

4.1 Modality-Agnostic Shared KV-Cache

The key insight of GPT-4o-style architectures OpenAI [2024] is that the Transformer’s key-value (KV) cache Pope et al. [2023] provides a natural mechanism for modality-agnostic state persistence.

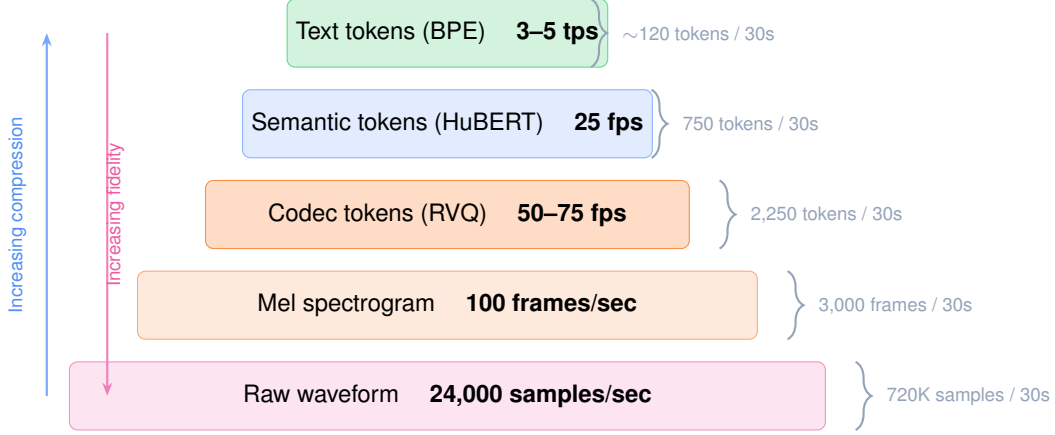


Figure 1: The **framerate pyramid** for audio representations. Each level compresses temporal resolution while abstracting toward semantic content. The substrate gap manifests as the $100\text{--}6,000\times$ ratio between the top (text) and bottom (waveform) layers. Native multimodal models must bridge at least two non-adjacent levels.

During autoregressive generation, the KV-cache stores the keys and values of all previously processed tokens:

$$\mathbf{K}_t = [\mathbf{k}_1, \dots, \mathbf{k}_t], \quad \mathbf{V}_t = [\mathbf{v}_1, \dots, \mathbf{v}_t] \quad (6)$$

where each $\mathbf{k}_i, \mathbf{v}_i$ is the projection of token i 's hidden state, regardless of whether token i originated from text, audio, or a special control token. When the model transitions from processing audio input to generating a text response, the KV-cache retains the full audio context—prosody, content, and speaker state—without requiring any explicit modality-switching mechanism.

This design achieves **zero-cost handoff**: the model does not need to encode, summarize, or re-embed one modality's context before operating in another. The Transformer's self-attention mechanism naturally cross-attends between cached audio keys and new text query vectors, enabling the model to reason about audio content in the same computation that produces text tokens.

4.2 Parallel Dual-Stream Decoding

Moshi Défossez et al. [2025] introduces an alternative: rather than sequential modality switches, it runs **two parallel generation streams** within a single Transformer backbone (Figure 2).

1. **Text stream** ("inner monologue"): Generates text tokens at standard LLM speed, representing the model's semantic reasoning. The text stream operates as a conventional language model, producing text tokens that are not directly rendered as audio but serve as a planning signal.
2. **Audio stream**: Generates codec tokens at the higher audio frame rate, conditioned on both the input audio and the text stream's outputs. A *delay pattern* offsets the audio stream by d timesteps behind the text stream:

$$p(a_t \mid a_{<t}, s_{<t+d}) = \text{Transformer}(\mathbf{h}_t^{\text{audio}} \mid \mathbf{KV}_{<t}^{\text{audio}}, \mathbf{KV}_{<t+d}^{\text{text}}) \quad (7)$$

where $s_{<t+d}$ denotes text tokens up to timestep $t + d$ and d typically corresponds to 1–2 text token positions (roughly 200–400 ms of lookahead).

This architecture supports **full-duplex conversation**: the model simultaneously processes incoming user audio and generates outgoing audio, with the text stream mediating between comprehension and production. The delay d is the critical hyperparameter—too small, and the text stream has not yet "decided" what to say; too large, and the perceived latency increases.

4.3 Continuous Integrate-and-Fire Streaming

For streaming applications where even 200 ms is significant, the **Continuous Integrate-and-Fire** (CIF) mechanism Dong and Xu [2019] enables sub-frame-level triggering. CIF maintains a running

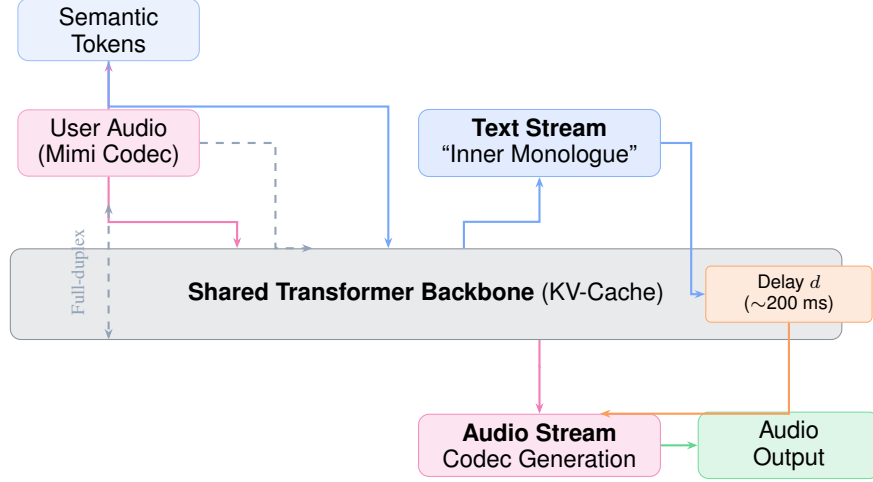


Figure 2: The **dual-stream decoding** architecture (Moshi-style). User audio enters the shared Transformer backbone as both raw codec tokens and extracted semantic tokens. The model generates two parallel output streams: a text “inner monologue” for semantic planning and an audio stream for speech output. The audio stream is delay-aligned behind the text stream by $d \approx 200$ ms, ensuring that audio generation is guided by semantic decisions without introducing perceptible latency.

accumulator α_t over input audio frames:

$$\alpha_t = \alpha_{t-1} + w_t, \quad \text{fire when } \alpha_t \geq \tau \quad (8)$$

where $w_t = \sigma(\mathbf{w}^\top \mathbf{h}_t + b)$ is a learned scalar weight for frame t and τ is the firing threshold (typically $\tau = 1.0$). When the accumulator fires, it emits a soft token and resets: $\alpha_t \leftarrow \alpha_t - \tau$. This produces a *variable-length* token sequence whose density adapts to speech content: rapid speech fires more frequently; silence barely fires at all.

The CIF mechanism enables **speculative reasoning**: the language model can begin processing partial acoustic input before the user finishes speaking, updating its internal state with each fired token. Combined with tree-based speculative decoding, this allows the model to prepare multiple candidate responses in parallel, selecting the appropriate one when the user’s utterance concludes.

5 Bridge IV: Preserving Paralinguistics

The three preceding bridges address the structural and temporal aspects of the substrate gap. The fourth addresses its most subtle dimension: *paralinguistic* information—the emotional, prosodic, and stylistic content that text fundamentally cannot encode but that is essential for natural conversation Schuller and Batliner [2013].

5.1 Interleaved Style Tokens

Spirit LM Nguyen et al. [2024] introduces *explicit paralinguistic tokens* interleaved within the text token stream. After each text segment, the model inserts discrete tokens encoding:

- **Pitch (F_0) tokens:** Quantized fundamental frequency, typically into 16–32 bins spanning the speaker’s range.
- **Style/tone tokens:** Discrete emotion or speaking-style categories (e.g., NEUTRAL, HAPPY, WHISPER, URGENT).

A typical interleaved sequence takes the form:

[TEXT] “I need help” [/TEXT]
[PITCH : HIGH] [RATE : FAST] [STYLE : FRANTIC]

This explicit encoding has a crucial advantage: the language model can *reason* about paralinguistic state using the same attention mechanism it uses for semantic content. If the model observes [STYLE: frantic] in the user’s input, it can condition its response on that state—choosing calming language, modulating its own pitch tokens, and adjusting speaking rate.

5.2 Persistent Style Vectors

An alternative to discrete style tokens is a *continuous style vector* that aggregates paralinguistic information from the user’s audio over time. The style vector $\mathbf{s} \in \mathbb{R}^{d_s}$ is computed by pooling over the KV-cache entries corresponding to user audio tokens:

$$\mathbf{s} = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \text{MLP}(\mathbf{v}_i) \quad (9)$$

where $\mathcal{A}$ is the set of positions corresponding to user audio tokens and $\mathbf{v}_i$ is the value vector at position i in a designated “style” attention layer. This vector is then injected into the audio decoder via cross-attention:

$$\mathbf{h}_t^{\text{out}} = \text{Attn}(\mathbf{h}_t^{\text{dec}}, \mathbf{K}_{\text{content}}) + \lambda \cdot \text{CrossAttn}(\mathbf{h}_t^{\text{dec}}, \mathbf{s}) \quad (10)$$

where λ controls the influence of the style vector. This mechanism enables **persistent style conditioning**: the model’s vocal output continuously reflects the user’s emotional state, even across long multi-turn conversations.

5.3 The Emotional Coherence Problem

A critical requirement for natural dialogue is *emotional coherence*: the model’s response should be contextually appropriate to the user’s emotional state. This is not merely a stylistic preference—it is a functional requirement. If a user whispers a frightened question, a model that responds in a loud, cheerful voice has failed at a fundamental level of communicative competence.

Formally, let e_{user} denote the detected user emotion and e_{model} the model’s output emotion. Emotional coherence requires a learned mapping $f : e_{\text{user}} \times c_{\text{semantic}} \rightarrow e_{\text{model}}$ that is *context-dependent*: the appropriate response to anger might be calm reassurance in a customer-service context but matched intensity in a debate context.

Current systems handle this with varying sophistication. GPT-4o OpenAI [2024] appears to use style conditioning vectors derived from prompt instructions. Moshi Défossez et al. [2025] relies on the dual-stream architecture’s implicit conditioning. Spirit LM Nguyen et al. [2024] makes the mechanism explicit through style tokens but does not model context-dependent emotion mapping. None of the current systems has demonstrated robust emotional coherence across diverse conversational scenarios, suggesting this remains an open and important problem.

6 Systems Analysis and Comparison

We now ground our taxonomy in a comparative analysis of five systems that implement different combinations of the four bridges. Table 2 provides a structured comparison.

Architectural lineage. These systems can be organized along a spectrum from “text-centric with audio adapters” to “audio-native with text emergence.” SpeechGPT and Spirit LM occupy the text-centric end: they augment a pretrained text LLM with audio tokens, preserving the sequential generation paradigm. Moshi occupies the audio-native end: it is designed from the ground up for parallel audio processing, with text emerging as an auxiliary stream. GPT-4o and GLM-4-Voice occupy intermediate positions, though architectural details for GPT-4o remain proprietary.

Latency characteristics. The handoff mechanism has the most direct impact on conversational latency. Systems using sequential modality switching (SpeechGPT, Spirit LM) incur at least one full encoding-decoding cycle at each transition, typically 500–1,000 ms. Moshi’s parallel dual-stream architecture achieves reported latency of ~ 200 ms Défossez et al. [2025] by eliminating sequential handoff entirely. GPT-4o achieves comparable latency through an undisclosed mechanism that likely involves shared KV-cache routing OpenAI [2024].

Table 2: Comparative analysis of native multimodal speech-language systems across the four substrate-gap bridges. “–” indicates unknown or not applicable.

System	Vocabulary (Bridge I)	Temporal (Bridge II)	Handoff (Bridge III)	Paralinguistics (Bridge IV)
GPT-4o OpenAI [2024]	Joint vocabulary (proprietary codec)	Proprietary downsampling	Shared backbone with unified KV- cache	Style conditioning via system prompt
Moshi Défossez et al. [2025]	Dual-stream RVQ (Mimi codec)	Semantic–acoustic factorization (HuBERT-aligned)	Parallel dual-stream decoding with delay alignment	Multi-stream implicit condition- ing
Spirit LM Nguyen et al. [2024]	Interleaved tokens (HuBERT codes + BPE text)	Fixed HuBERT downsampling (25 Hz)	Sequential with modality markers	Explicit F_0 and style tokens
GLM-4-Voice Zeng et al. [2024]	Whisper encoder + CosyVoice token- izer	CosyVoice learned compression	End-to-end fine-tuning	Voice cloning via CosyVoice decoder
SpeechGPT Zhang et al. [2023]	Discrete HuBERT codes in LLM vo- cab	Fixed downsam- pling (HuBERT 50 Hz → 25 Hz)	Sequential (ASR → LM → TTS)	Limited (no explicit mecha- nism)

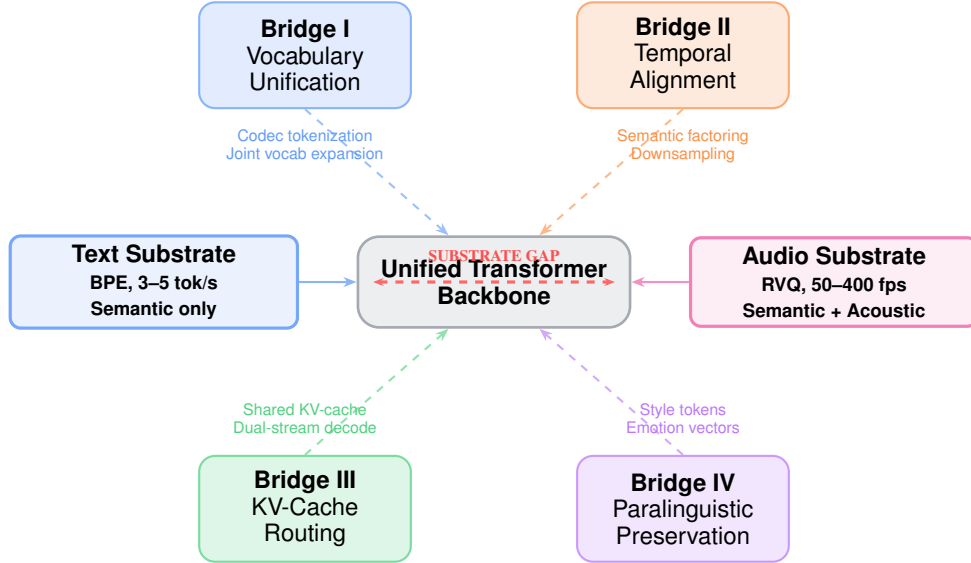


Figure 3: Architectural overview of the four bridges that dissolve the substrate gap between text and audio modalities. Each bridge addresses a specific dimension of the mismatch: representational (*Bridge I*), temporal (*Bridge II*), computational (*Bridge III*), and contextual (*Bridge IV*). A unified Transformer backbone integrates all four bridges to enable native multimodal processing.

Paralinguistic fidelity. Spirit LM’s explicit style tokens provide the most interpretable paralinguistic mechanism, at the cost of a coarse-grained discretization. Moshi’s multi-stream approach implicitly preserves paralinguistic information through the acoustic RVQ levels but does not provide explicit access for reasoning. GLM-4-Voice delegates paralinguistic generation to CosyVoice Du et al. [2024], achieving high vocal fidelity but without bidirectional paralinguistic reasoning (the model does not condition its text response on the user’s emotional state).

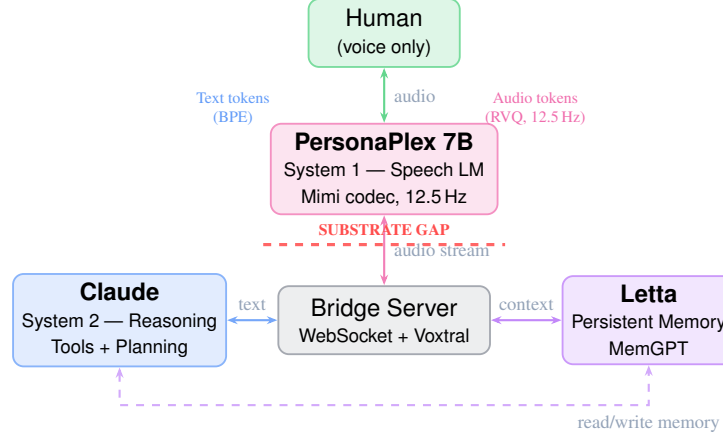


Figure 4: The **sidecar architecture** for bridging the substrate gap without a unified model. PersonaPlex (System 1) owns the human relationship via full-duplex audio. Claude (System 2) provides reasoning but never speaks to the user directly. The substrate gap manifests at the Bridge Server, which must translate between audio-token and text-token substrates. Letta provides persistent memory shared by both systems.

7 Case Study: The Sidecar Architecture

The preceding analysis describes systems that dissolve the substrate gap *within* a single model. An alternative—motivated by practical constraints where no single model yet achieves all four bridges—is the **sidecar architecture**: a multi-agent system in which a Speech Language Model (Speech LM) handles real-time audio interaction while a separate reasoning LLM operates “backstage,” communicating through a mediation layer. We present an implemented system that exposes the substrate gap as a concrete engineering problem rather than a theoretical construct.

7.1 Architecture

The system comprises four components organized in a dual-process cognitive architecture inspired by Kahneman’s System 1/System 2 framework Kahneman [2011]:

1. **PersonaPlex** (System 1 / Voice Interface): NVIDIA’s fine-tune of Moshi NVIDIA et al. [2025], a 7B-parameter full-duplex speech-text foundation model running on a consumer GPU (NVIDIA RTX 3090, 24GB). PersonaPlex handles real-time conversation at 12.5 Hz via the Mimi codec, maintaining the Moshi dual-stream architecture (§4.2) with an added *text prompt* mechanism for persona conditioning and a *voice prompt* for speaker identity. PersonaPlex is the *sole interface* to the human user—the only voice they hear.
2. **Claude** (System 2 / Reasoning Engine): A frontier text-based LLM Anthropic [2025] that handles complex reasoning, tool calling (web search, database queries, code execution), and multi-step planning. System 2 never communicates directly with the user.
3. **Letta** (Persistent Memory): A stateful memory agent implementing the MemGPT architecture Packer et al. [2024] with structured memory blocks (persona, human profile, conversation buffer) that persist across sessions. Both System 1 and System 2 read and write to Letta, providing cross-session continuity.
4. **Bridge Server** (Mediator): A WebSocket server that routes audio between the user’s browser and PersonaPlex, transcribes speech via Voxtral Mistral AI [2024] for System 2 consumption, manages turn-taking, and delivers System 2 responses back to the user.

Figure 4 illustrates the information flow. The key design constraint is that the user should perceive a single, unified agent—“Pluribus mode”—despite the system comprising four separate components.

7.2 The Content Injection Problem

The central engineering challenge is **content injection**: how does System 2’s text-based reasoning output reach the user through System 1’s audio-native interface?

PersonaPlex’s `text_prompt` mechanism was designed for *persona conditioning*—setting behavioral attributes such as role, speaking style, and background knowledge. Empirical testing revealed that it *cannot* reliably deliver dynamic content. Three injection strategies were tested:

1. **Direct content embedding**: Appending System 2’s answer directly to the text prompt (“Tell the user: The answer is 42”). *Result*: PersonaPlex acknowledged the information existed but paraphrased, hallucinated details, or ignored it entirely.
2. **Structured framing**: Formatting the answer as persona-relevant knowledge (“Your research team has confirmed that...”). *Result*: Marginal improvement for simple facts (NVIDIA reports 4.48/5 factual delivery for service scenarios NVIDIA et al. [2025]), but complex multi-sentence answers were consistently corrupted.
3. **Session reconnect**: Disconnecting and reconnecting with an updated text prompt containing fresh content. *Result*: 2–3 second interruption, no guarantee of content delivery, and loss of conversational state.

This negative result is architecturally informative. PersonaPlex (and Moshi more broadly) processes text prompts as a *conditioning signal* during the initial timesteps of generation, analogous to a classifier-free guidance vector. The text tokens are consumed during prefilling and influence the KV-cache but are not “remembered” as retrievable instructions. This is a direct consequence of the dual-stream architecture (§4.2): the inner monologue stream is the model’s *own* text prediction, not an instruction-following channel.

7.3 The Provided-Flag Token Override

Analysis of PersonaPlex’s inference code reveals a mechanism that appears capable of resolving the injection problem without fine-tuning. During autoregressive generation, the model maintains a provided boolean flag per timestep that determines whether to use an externally supplied token or the model’s own sampled token:

$$t_{\text{next}} = \begin{cases} t_{\text{injected}} & \text{if provided}[t] = \text{True} \\ t_{\text{sampled}} & \text{otherwise} \end{cases} \quad (11)$$

This mechanism is used during system prompt prefilling, where the first N text tokens are force-fed before free generation begins. There is no architectural constraint limiting its use to the initial timesteps, suggesting it could enable mid-conversation content injection.

We implemented this approach (the “Pluribus Earpiece”): System 2’s text responses are tokenized via PersonaPlex’s sentencepiece tokenizer, queued, and dripped into the `step()` function at one token per audio frame (~ 12.5 tokens/sec), matching natural speech rate. The injection is **functionally successful**: PersonaPlex accepts the tokens, and the Depformer generates audio output corresponding to the injected text content. However, the resulting audio is **severely garbled**—intelligible in fragments but acoustically degraded, with artifacts including pitch discontinuities, phoneme distortion, and unnatural prosody.

7.4 Root Cause Analysis: The Conditioning Mismatch

The garbled audio stems from three compounding mechanisms inherent to the dual-stream architecture:

KV-cache trajectory divergence. The Temporal Transformer (7B backbone) computes its output $\mathbf{h}_t^{\text{out}}$ via self-attention over the KV-cache, which encodes the model’s *expected* continuation trajectory. When the provided flag overrides the sampled text token, the replacement occurs *after* the Temporal Transformer’s forward pass. The Depformer therefore receives contradictory conditioning signals:

$$\mathbf{d}_t = \underbrace{f_{\text{emb}}(t_{\text{injected}})}_{\text{injected token}} + \underbrace{\mathbf{h}_t^{\text{out}}}_{\substack{\text{computed assuming} \\ t_{\text{sampled}}}} \quad (12)$$

The additive combination of a foreign text embedding with a contextually inconsistent transformer output produces an out-of-distribution input to the Depformer, which has never encountered such combinations during training.

8× error amplification. Each text token conditions $Q = 8$ successive audio codebook predictions via the Depformer’s autoregressive chain. A single misaligned text token cascades through all 8 RVQ levels, amplifying the distributional error at each stage. This is consistent with the exposure bias literature, where error accumulation in autoregressive models is proportional to sequence depth He and Zhang [2019]. In audio generation, exposure bias is particularly damaging because the high density of acoustic tokens ($8\times$ per text token) provides more steps for errors to compound Zhang et al. [2025].

Delay pattern violation. Moshi’s dialogue mode uses a delay offset $d = 1$ between text and audio streams (Eq. 7), providing minimal lookahead. In contrast, Moshi’s proper TTS mode uses $d = 16$ (~ 1.28 seconds), giving the Depformer sufficient context to plan prosody and phoneme transitions Défossez et al. [2025]. Mid-conversation injection operates at dialogue-mode delays, denying the Depformer the temporal lookahead it needs for coherent speech planning.

Key finding. The provided flag was designed exclusively for *pre-conversation* prompt prefilling, where the KV-cache is empty and text tokens establish the initial trajectory rather than disrupting an existing one. This negative result confirms that Bridge III cannot be achieved by simple token substitution in dual-stream architectures; the handoff mechanism must respect the model’s internal state dynamics.

7.5 Viable Alternatives for Content Delivery

The empirical failure of mid-conversation token injection motivates three alternative Bridge III strategies, ranked by feasibility:

1. Dedicated TTS model (cascaded). Kyutai has released a standalone TTS model Kyutai [2025] built on the same Delayed Streams framework as Moshi but trained specifically for text-to-speech with a $d = 16$ delay pattern. System 2’s text is synthesized by this dedicated model and injected into the user’s audio stream, bypassing PersonaPlex entirely for factual content delivery. Voice cloning from PersonaPlex’s voice prompt can minimize the perceptual seam. *Expected latency:* +200 ms (streaming synthesis begins before full text is available).

2. Response boundary restart (GPT-4o pattern). Rather than injecting tokens mid-stream, the system ends the current generation turn, inserts System 2’s response as new conversational context, and initiates a fresh generation turn with a clean KV-cache. This mirrors how GPT-4o’s Realtime API handles tool call results: text is inserted as a `function_call_output` conversation item, and a new response is generated from consistent state OpenAI [2024]. *Expected latency:* +500 ms gap during re-initialization.

3. TTS mode switching. PersonaPlex can be temporarily switched from dialogue mode ($d = 1$) to TTS mode ($d = 16$), with all System 2 text tokens provided in advance. In this configuration, the Temporal Transformer processes text tokens during its forward pass and builds KV-cache entries *consistent* with the provided text—eliminating the conditioning mismatch of Eq. 12. *Expected latency:* +1.28s lookahead; requires mode-switching logic.

7.6 Current Deployed Workaround

While the above alternatives are under development, the deployed system uses a **browser TTS sidecar**: System 2’s text responses are synthesized via the browser’s `SpeechSynthesis` API and played through a separate audio channel while PersonaPlex’s output is temporarily muted.

This workaround is functional but violates the Pluribus design constraint: the user hears two distinct voices—PersonaPlex’s neural voice for conversation and the browser’s robotic TTS voice for factual answers. The substrate gap is not bridged but *bypassed*, with the seam clearly audible.

7.7 Implications for the Taxonomy

The sidecar architecture—and particularly the failure of mid-conversation token injection—reveals that the four bridges of our taxonomy interact in ways not apparent from the single-model perspective:

- **Bridge I** (Vocabulary): Voxtral transcription provides a lossy text-to-audio vocabulary mapping. The provided flag operates within PersonaPlex’s own vocabulary but fails because vocabulary alignment alone is insufficient without state consistency.
- **Bridge II** (Temporal): The delay pattern ($d = 1$ vs $d = 16$) is not merely a latency parameter but a *structural requirement* for coherent cross-modal generation. Mid-conversation injection fails Bridge II because it uses dialogue-mode delays for a TTS-mode task.
- **Bridge III** (Handoff): Token substitution is not a valid handoff mechanism when the model’s hidden state is inconsistent with the substituted tokens. Valid handoffs require either state re-computation (response boundary restart) or dedicated models trained for the target task (TTS model).
- **Bridge IV** (Paralinguistics): The garbled audio demonstrates that paralinguistic preservation depends on all other bridges being correctly implemented. An acoustically degraded delivery fails Bridge IV regardless of voice identity preservation.

The most significant finding is that **Bridge III cannot be decomposed from Bridge II**: the handoff mechanism and temporal alignment are coupled through the KV-cache. This coupling is invisible in single-model systems (where both are handled implicitly) but becomes a hard constraint in multi-agent architectures.

This case study suggests that the substrate gap, when bridged across system boundaries, requires either *dedicated bridging models* (trained specifically for cross-substrate translation) or *state-coherent injection* (where the receiving model’s internal state is consistent with the injected content). Simple token substitution—despite being mechanically possible—produces acoustically unacceptable results. As unified models like Voila Maitrix.org et al. [2025] mature—achieving 195 ms latency with integrated reasoning and voice—the sidecar architecture provides a migration path: components can be incrementally replaced by unified-model capabilities without redesigning the entire system.

8 Open Problems and Future Directions

Despite significant progress, several fundamental challenges remain unresolved. We organize these into five categories.

8.1 Multilingual Codec Transfer

Current neural codecs are predominantly trained on English-centric corpora. Transferring codebook representations to typologically diverse languages—including tonal languages (Mandarin, Yoruba), agglutinative languages (Turkish, Finnish), and low-resource languages—is largely unexplored. Tonal languages present a particular challenge because they use F_0 contours for *lexical* meaning, not just paralinguistic expression. A codec’s semantic-acoustic factorization may conflate lexical tone with prosodic emotion, producing catastrophic errors in tonal language processing.

The potential solution involves **language-adaptive codebook fine-tuning**: freezing the encoder architecture while fine-tuning codebook vectors on target-language data, possibly with language-specific first-level codebooks that capture language-dependent phonetic inventories while sharing higher-level acoustic codebooks across languages.

8.2 Real-Time Emotion Adaptation

Current systems detect user emotion at a single point in time, but emotions evolve continuously during conversation. A user may begin calmly, grow frustrated, and then relax after receiving help—all within a single conversational turn. Real-time emotion adaptation requires:

1. **Continuous emotion tracking**: An online estimator $e_t = f(\mathbf{h}_t, e_{t-1})$ that updates at every audio frame, not just at utterance boundaries.

2. **Emotion-conditioned generation:** The model’s output style must adapt mid-utterance to changing user emotion, requiring the style vector (Eq. 9) to be computed over a sliding window rather than a full utterance.
3. **Emotional memory:** The model should remember the emotional trajectory of a conversation, not just the current state, to respond appropriately to patterns (“you’ve been frustrated several times—let me try a different approach”).

8.3 Multi-Speaker Scene Understanding

All current native multimodal systems assume a single-user, single-model conversational dyad. Real-world scenarios frequently involve multiple speakers: meetings, classrooms, family interactions. Extending the substrate gap framework to multi-speaker settings requires:

- **Speaker-tagged codebooks:** Associating each audio token with a speaker embedding, enabling the model to track who said what.
- **Overlap handling:** Natural conversation involves frequent overlapping speech. The model must process simultaneous audio streams from multiple speakers, potentially using separate KV-cache partitions per speaker.
- **Addressee detection:** Determining whom the model should respond to, and when, in a multi-party setting.

8.4 Codec-Free End-to-End Approaches

Neural codecs introduce a discretization bottleneck: information lost during quantization cannot be recovered. An emerging research direction explores **codec-free** models that operate directly on continuous audio representations—spectrograms, learned embeddings, or even raw waveforms—within the Transformer backbone.

The challenge is computational: raw 24 kHz audio produces 24,000 values per second, far beyond current Transformer capacity. Potential solutions include learned downsampling within the model (essentially making the codec part of the model rather than a preprocessing step) and state-space models (SSMs) that can process long sequences more efficiently than attention-based Transformers.

8.5 Ethical Considerations: Voice Cloning and Deepfakes

The same technology that enables natural voice generation creates unprecedented risks for misuse Khanjani et al. [2023], Yi et al. [2024]. Systems like VALL-E Wang et al. [2023] can clone a speaker’s voice from 3 seconds of audio. As native multimodal models become more capable, several concerns intensify:

- **Non-consensual voice synthesis:** A model that has processed a user’s audio can potentially reproduce their voice, raising consent and intellectual property questions.
- **Real-time impersonation:** Sub-200 ms latency enables live voice impersonation during phone calls or video conferences.
- **Paralinguistic manipulation:** A model that understands and generates emotion could be used to manipulate users’ emotional states through carefully crafted prosodic patterns.
- **Detection arms race:** Audio deepfake detection Yi et al. [2024] is increasingly difficult as generation quality improves, particularly for codec-based generation that does not exhibit the spectral artifacts of older TTS systems.

We advocate for **provenance-aware generation:** embedding cryptographic watermarks in generated audio at the codec level, before decoding to waveforms. This is architecturally natural for discrete-codec systems, where watermark tokens can be interleaved with content tokens, but remains an open problem for continuous-representation systems.

9 Conclusion

The substrate gap between text and audio tokens is not a single problem but a composite of four interacting mismatches: representational, temporal, computational, and contextual. We have surveyed the architectural patterns—discrete neural codecs, semantic–acoustic factorization, shared KV-cache routing, and interleaved style tokens—that collectively address these mismatches and enable native real-time multimodal intelligence.

Our analysis reveals that no current system fully resolves all four dimensions simultaneously. Moshi achieves the most complete integration through its dual-stream architecture and Mimi codec, but sacrifices explicit paralinguistic reasoning. Spirit LM provides the most transparent paralinguistic mechanism but uses sequential handoff with higher latency. GPT-4o may address all four dimensions, but its proprietary architecture prevents independent verification. Our case study of the PersonaPlex sidecar architecture demonstrates that the four bridges can be addressed *across system boundaries* through multi-agent orchestration, with the provided-flag token override offering a promising zero-training path for content injection into Speech LMs.

The field is converging toward a common architectural template: a single Transformer backbone with a joint discrete vocabulary spanning both modalities, semantic–acoustic factorization for temporal alignment, a shared KV-cache for zero-cost modality handoff, and some form of style conditioning for paralinguistic preservation. The remaining open problems—multilingual transfer, real-time emotion adaptation, multi-speaker understanding, and ethical safeguards—define the research frontier for the next generation of conversational AI systems.

As these systems mature, the substrate gap will not merely be bridged but dissolved: future models may operate on a single, unified token space where the distinction between text and audio is a training-time annotation rather than an architectural boundary.

Acknowledgments and Disclosure of Funding

The author thanks the open-source research community, particularly the teams behind Moshi (Kyutai), Spirit LM (Meta FAIR), and AudioLM (Google DeepMind), whose published architectures and codebases made this analysis possible. This work was conducted at Viable Systems Research and received no external funding.

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A Notation Summary

Table 3: Summary of mathematical notation used throughout the paper.

Symbol	Description
$\mathbf{x} \in \mathbb{R}^{T \cdot f_s}$	Raw audio waveform (T seconds at sample rate f_s)
$E(\cdot)$	Neural codec encoder
$\mathbf{z} \in \mathbb{R}^{T' \times d}$	Latent audio representation
$VQ_i(\cdot)$	Vector quantizer at RVQ stage i
$\mathcal{C}_i$	Codebook for RVQ stage i , $ \mathcal{C}_i = K$
K	Codebook size (typically 1,024)
Q	Number of RVQ levels
$\mathcal{V}_{\text{text}}$	Text vocabulary (BPE)
$\mathcal{V}_{\text{audio}}$	Audio token vocabulary
$\mathcal{V}_{\text{joint}}$	Joint multimodal vocabulary
d_{model}	Transformer hidden dimension
$\mathbf{K}_t, \mathbf{V}_t$	KV-cache at timestep t
α_t	CIF accumulator value at frame t
w_t	CIF learned weight for frame t
τ	CIF firing threshold
d	Delay offset (dual-stream)
$\mathbf{s}$	Persistent style vector
$e_{\text{user}}, e_{\text{model}}$	User / model emotional state